

MA 301

Test 3

Fall - 2014

Problem	points	out of
1		8
2		8
3		8
EC		2
Score		24

Name: SOLUTIONS

HR: _____

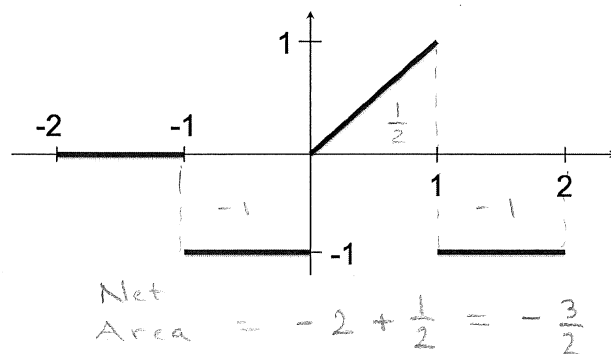
Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.

$$d_0 = \frac{1}{4} \overbrace{\int_{-2}^2 f(x) dx}^{\text{AREA}} = \frac{1}{4} \left(-\frac{3}{2} \right) = -\frac{3}{8}$$



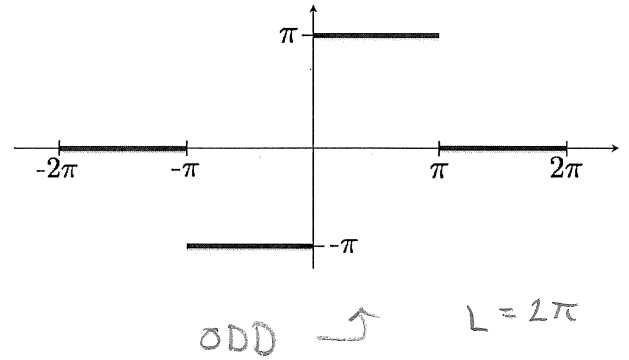
$$d_n = \frac{1}{4} \int_{-2}^2 f(x) e^{-i\frac{\pi}{2}nx} dx = \frac{1}{4} \left(\int_{-1}^0 (-1) e^{-i\frac{\pi}{2}nx} dx + \int_0^1 x e^{-i\frac{\pi}{2}nx} dx + \int_1^2 (-1) e^{-i\frac{\pi}{2}nx} dx \right)$$

SET-UP ONLY

$$f(x) = -\frac{3}{8} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} d_n e^{i\frac{\pi}{2}nx}$$

where d_n would come from the computation above.

Problem 2: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.



$$d_0 = 0$$

$$d_n = \frac{-i}{2\pi} \int_0^{2\pi} f(x) \sin\left(\frac{n}{2}x\right) dx$$

$$= -\frac{i}{2\pi} \int_0^{\pi} \pi \sin\left(\frac{n}{2}x\right) dx$$

$$= -\frac{i}{2} \left(-\frac{2}{n} \cos\left(\frac{n}{2}x\right) \right) \Big|_0^{\pi}$$

$$= \frac{i}{n} \left(\cos\left(\frac{\pi}{2}n\right) - \cos(0) \right) = \frac{i}{n} \left(\cos\left(\frac{\pi}{2}n\right) - 1 \right)$$

$$f(x) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} d_n e^{i\frac{n}{2}x}$$

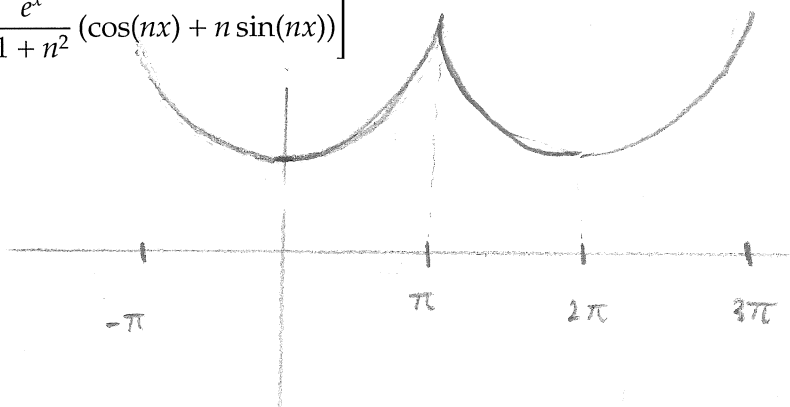
Problem 3: Consider the periodic function $f(x) = \begin{cases} e^{-x} & -\pi \leq x \leq 0 \\ e^x & 0 < x \leq \pi \end{cases}$ and $f(x) = f(x + 2\pi)$

(a) Plot $f(x)$ on the interval $[-\pi, 3\pi]$ (yes, I meant 3π)

(b) Find the Fourier Series of $f(x)$

[Hint: An anti-derivative of $e^x \cos(nx)$ is $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$]

(a)



Even

$L = \pi$

$$(b) \quad d_0 = \frac{1}{\pi} \int_0^{\pi} e^x dx = \frac{1}{\pi} e^x \Big|_0^{\pi} = \frac{1}{\pi} (e^{\pi} - 1)$$

$$d_n = \frac{1}{\pi} \int_0^{\pi} e^x \cos(nx) dx = \frac{1}{\pi} \left(\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx)) \right) \Big|_0^{\pi}$$

$$= \frac{1}{\pi} \left(\frac{e^{\pi}}{1+n^2} (-1)^n - \frac{1}{1+n^2} (1) \right)$$

$$= \frac{e^{\pi}}{\pi(1+n^2)} ((-1)^n - 1)$$

$$f(x) = \frac{e^{\pi} - 1}{\pi} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{e^{\pi}}{\pi(1+n^2)} ((-1)^n - 1) e^{inx}$$

Extra Credit: Show that $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$ is an anti-derivative of $e^x \cos(nx)$

Note: Do not just take the derivative of $\frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$

$$\int e^x \cos(nx) dx$$

$$u = \cos(nx) \quad v = e^x$$
$$du = -n \sin(nx) \quad dv = e^x dx$$

$$= e^x \cos(nx) + n \int e^x \sin(nx) dx$$

$$u = \sin(nx) \quad v = e^x$$
$$du = n \cos(nx) \quad dv = e^x dx$$

$$= e^x \cos(nx) + n \left[e^x \sin(nx) - n \int e^x \cos(nx) dx \right]$$

$$= e^x \cos(nx) + n e^x \sin(nx) - \underbrace{n^2 \int e^x \cos(nx) dx}_{**}$$

$$\Rightarrow \underbrace{\int e^x \cos(nx) dx}_{*} + \underbrace{n^2 \int e^x \cos(nx) dx}_{**} = e^x (\cos(nx) + n \sin(nx))$$

$$\Rightarrow (1+n^2) \int e^x \cos(nx) dx = e^x (\cos(nx) + n \sin(nx))$$

$$\Rightarrow \int e^x \cos(nx) dx = \frac{e^x}{1+n^2} (\cos(nx) + n \sin(nx))$$